

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Audio Processor
Manufacturer: Apart
Firmware Version: N/A
Model(s): Concept 1

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.05.0000-b007	2.01.0002-b005	1.3.0

Version History

Module Version	Date	Notes
1_4_0_0	6/13/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5
- For Serial Over Ethernet support, SerialOverEthernetClass must be used. Example: InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', PortNum, Model='ModelName')
- The mode of Concept 1 is determined by MULTIZONE and ZONELINK. If MULTIZONE is OFF, the amplifier is working in stereo mode. If MULTIZONE is ON, the amplifier will work in 2 zone mode. If ZONELINK is ON, the music level of zone 1 and zone 2 are linked. Otherwise, when ZONELINK is OFF, the 2 zones have a separate volume setting. Since these 2 commands change the behavior of the amplifier, it's recommended only to change them when the amplifier is not in action (no music present, volume at minimum). If the amplifier needs to be configured in another mode, these are the first settings that need to be changed.
- When Multizone is 'On' only qualifier "1 or 2" will work otherwise Zone qualifier should be "Multizone Off". Multizone and Zonelink default state is 'Off'.

Control Commands, States, and Qualifiers:

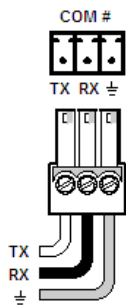
AutoLevel	'On'	'Off'
Bass	-14 to 14 in steps of 2	
Input	'A'	'B' 'C'
	'D'	
InputGain	-20 to 14 in steps of 2	
['Input']	'A'	'B' 'C'
	'D'	
Multizone	'On'	'Off'
PagingActive	'On'	'Off'
['Zone']	'1'	'2' 'Multizone Off'
Power	'On'	'Off'
Treble	-14 to 14 in steps of 2	
ZoneMicLevel	-80 to 0 in steps of 1	
['Zone']	'1'	'2' 'Multizone Off'
ZoneMusicLevel	-80 to 0 in steps of 1	
['Zone']	'1'	'2' 'Multizone Off'
Zonelink	'On'	'Off'

Status Available:

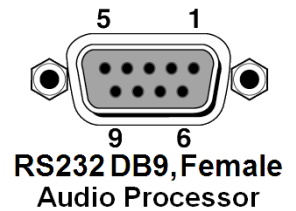
AutoLevel	'On'	'Off'
Bass	-14 to 14 in steps of 2	
ConnectionStatus	'Connected'	'Disconnected'
Input	'A'	'B' 'C'
	'D'	
InputGain	-20 to 14 in steps of 2	
['Input']	'A'	'B' 'C'
	'D'	
Multizone	'On'	'Off'
Paging	'On'	'Off'
['Zone']	'1'	'2' 'Multizone Off'
PagingActive	'On'	'Off'
['Zone']	'1'	'2' 'Multizone Off'
Treble	-14 to 14 in steps of 2	
ZoneMicLevel	-80 to 0 in steps of 1	
['Zone']	'1'	'2' 'Multizone Off'
ZoneMusicLevel	-80 to 0 in steps of 1	
['Zone']	'1'	'2' 'Multizone Off'
Zonelink	'On'	'Off'

Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device:**Serial communication:****Port Type:** RS-232**Baud Rate:** 19200**Data Bits:** 8**Parity:** None**Stop Bits:** One**Flow Control:** None**Pin Assignments Diagram**

Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND

**General Notes:**

Appendix A. Set Commands

Auto Level Off	SET AUTOLD 0\x0D
Auto Level On	SET AUTOLD 1\x0D
Bass 14	SET EQBASS 14\x0D
Bass -14	SET EQBASS -14\x0D
Input A	SET SELECT A\x0D
Input B	SET SELECT B\x0D
Input C	SET SELECT C\x0D
Input D	SET SELECT D\x0D
Input Gain 14 Input A	SET IPGAIN A 14\x0D
Input Gain 14 Input B	SET IPGAIN B 14\x0D
Input Gain 14 Input C	SET IPGAIN C 14\x0D
Input Gain 14 Input D	SET IPGAIN D 14\x0D
Input Gain -20 Input A	SET IPGAIN A -20\x0D
Input Gain -20 Input B	SET IPGAIN B -20\x0D
Input Gain -20 Input C	SET IPGAIN C -20\x0D
Input Gain -20 Input D	SET IPGAIN D -20\x0D
Multizone Off	SET MULTIZONE 0\x0D
Multizone On	SET MULTIZONE 1\x0D
Paging Active Off Zone Multizone Off	SET PAGACT 0\x0D
Paging Active On Zone Multizone Off	SET PAGACT 1\x0D
Power Off	SET STANDBY 1\x0D
Power On	SET STANDBY 0\x0D
Treble 14	SET EQTREB 14\x0D
Treble -14	SET EQTREB -14\x0D
Zone Mic Level 0 Zone Multizone Off	SET MICLVL 0\x0D
Zone Mic Level -80 Zone Multizone Off	SET MICLVL -80\x0D
Zone Music Level 0 Zone Multizone Off	SET MSCLVL 0\x0D
Zone Music Level -80 Zone Multizone Off	SET MSCLVL -80\x0D
Zonelink Off	SET ZONELINK 0\x0D
Zonelink On	SET ZONELINK 1\x0D

Appendix B. Query Commands

Auto Level	GET AUTOLD\x0D
Bass	GET EQBASS\x0D
Input	GET SELECT\x0D
Input Gain Input A	GET IPGAIN A\x0D
Input Gain Input B	GET IPGAIN B\x0D
Input Gain Input C	GET IPGAIN C\x0D
Input Gain Input D	GET IPGAIN D\x0D
Multizone	GET MULTIZONE\x0D
Paging Active Zone Multizone Off	GET PAGACT\x0D
Treble	GET EQTREB\x0D
Zone Mic Level Zone Multizone Off	GET MICLVL\x0D
Zone Music Level Zone Multizone Off	GET MSCLVL\x0D
Zonelink	GET ZONELINK\x0D